EXTENSION OF ALGEBRAICALLY CLOSED FIELDS DOES NOT CHANGE THE KERNEL OF THE FROBENIUS LINEAR MAP MINUS ONE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the coordinate description of the kernel is written out, the finiteness input is attributed to the companion lemma, and the final base change step is stated.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The situation and conclusion of Lemma F-1. Same chapter, Lemma lemma-F-1: in the standing situation the equations cutting out the kernel define a closed subscheme of affine space finite over Spec(k). Supplies both the situation the statement refers to and the finiteness used in the proof.
- Points of a finite scheme over an algebraically closed field after an extension. Standard base change behaviour of the k-valued points of a scheme finite over Spec(k) when k is algebraically closed, as used by the author in the closing sentence. Supplies the equality Z(k) = Z(k') that finishes the proof.

Lemma 0.1. *Let k be an algebraically closed field of characteristic $p > 0$. Let V be a finite dimensional k-vector space. Let $F : V \to V$ be a frobenius linear map, i.e., an additive map such that $F(\lambda v) = \lambda^p F(v)$ for all $\lambda \in k$ and $v \in V$. Then $F - 1 : V \to V$ is surjective with kernel a finite dimensional $\mathbf{F}_p$-vector space of dimension $\leq \dim_k(V)$.*

Lemma 0.2. *In the situation of Lemma 0.1 let k'/k be an extension of algebraically closed fields. Set $V' = V \otimes_k k'$ and $F' : V' \to V'$ be the Frobenius linear map induced by F. Then $\mathrm{Ker}(F - 1)$ maps isomorphically onto $\mathrm{Ker}(F' - 1)$.*

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Proof. Let $v_1, \ldots, v_n$ be a basis of V and say $F(v_i) = \sum a_{ij} v_j$. Then $v = \sum \lambda_i v_i$ is in $\mathrm{Ker}(F - 1)$ if and only if $\sum \lambda_i^p a_{ij} = \lambda_j$ for $j = 1, \ldots, n$. By Lemma 0.1 these equations cut out a closed subscheme Z of $\mathbf{A}^n_k = \mathrm{Spec}(k[\lambda_1, \ldots, \lambda_n])$ finite over $\mathrm{Spec}(k)$. Then $Z(k) = Z(k')$ and we conclude. $\square$